

Network Remote Identification Service Description Document

General Service Description	
Service description identification	Title: Network Remote Identification Service Description Edition: V1.1.0 Reference Date: Sep 24, 2026
Service identification	Name: ASTM F3411-22a Network Remote Identification Service
Service abstract	This ASTM F3411-22a standard-based network remote identification service is aimed at meeting UAS identification, situational awareness, and counter UAS needs, and as a proposed alternate means for an operator to meet §108.200(4) Range of broadcast operator requirements or equivalent.
Service Characteristics	
Service standard reference	Service standard reference: Service Standard Reference: Standard Specification for Remote ID and Tracking (ASTM F3411-22a) Implemented options of the service standard: • Net-RID Service Providers and Net-RID Display Providers must provide the following optional fields in ASTM F3411-22a Table 1 as required under §108.200(2) and §89.305: ◦ UAS ID - 1. Serial Number or 3. UTM (UUID) ◦ Operator Latitude ◦ Operator Longitude ◦ Operator Altitude ◦ Operator Location Type ◦ Operational Status • Net-RID Service Providers shall retain data for a minimum period of 45 days. • The following Display Provider requirements are not applicable to Display Providers only when providing data for meeting Counter UAS needs: ◦ NET0480 ◦ NET0490
High-level Description of Service Offer	
Operational environment	Refer to ASTM F3411-22a Sections 4.5-4.8 for operational needs addressed by the service and the capabilities offered by the service.

Service functions	Refer to the following ASTM F3411-22a sections for functions offered by the service and their real world effects. Section 5.5, 10.3, A2
Limitations and Constraints on Using the Service	
Service access and use conditions	Refer to the applicable UAS Service Provider Data Sharing and Governance Agreement and Operator - Service Provider Agreement
Security constraints	Refer to the following documents for security requirements. • ASTM F3411-22a Sections 5.5.4 and A2: ○ NET0210 ○ NET0220 ○ DSS0010 ○ DSS0020 ○ DSS0110 ○ DSS0120 • ASTM F3548-21 Sections 5.2 and A2: ○ GEN0005 - USSs performing any of the roles identified in this specification shall be implemented and operated under an ISO/IEC 27001-compliant Information Security Management System or equivalent. Service Provider may demonstrate equivalent compliance to ISO/IEC 27001 by providing a written attestation that the Service Provider materially complies with ISO/IEC 27001 or has implemented an Information Security Management System that includes the following: • Risk Assessment. Annual information security risk assessment of UTM systems. • Access Control. The Service Provider will limit access to its UTM Service and Shared Data solely to Service Provider employees and contractors who need access to perform their duties. The Service Provider has sufficient controls in place to prevent unauthorized access to its UTM Services and Shared Data. At a minimum such controls will include multi-factor authentication, periodic review of access privileges, and immediate revocation of access privileges upon an employee's termination. • Data Control. Shared Data obtained in the operation of UTM Services is encrypted in transit. Shared Data is stored only on systems controlled by the Service Provider or via security-certified vendors. • Monitoring. Systems used in providing the UTM Services are monitored for configuration drift and unauthorized access. All employee endpoints run malware screening and remediation tooling. • Incident Response. Established process for incident response and breach notification.

Privacy	Refer to ASTM F3548-21 Sections 5.2 and A2 for privacy requirements. Relevant requirements are listed below: * GEN0010 - USSs performing any of the roles identified in this specification shall be implemented and operated under an ISO/IEC 27701-compliant Privacy Information Management System or equivalent. For initial production flights, Service Providers may demonstrate equivalent compliance for ISO/IEC 27701 as required by ASTM F3548-21 as follows: • A Service Provider conforms with the terms set forth in Annex D (Data Privacy) • A Service Provider completes an annual self-assessment and makes a written attestation during onboarding that the Service Provider materially complies or has implemented the following: ○ Conduct annual privacy risk assessment. ○ Implement privacy controls as part of a Privacy Information Management System (PIMS). ○ Implement privacy and security roles and responsibilities. ○ Define and adhere to a consent management process. ○ Manage Data Subject Rights (including as set forth in Annex D). ○ Establish processes for incident response and breach notification.
Authentication	Per ASTM F3548-21, Appendix X1 - Reference architecture for interoperability security controls, The X1.1 Base Deployment: Access Tokens with Audience Claims is implemented. Authentication and Authorization for communication between USS-USS and USS-DSS shall be facilitated by a common interoperability token exchanger service supporting OAuth 2.0. Auth provider should be ISO27001 certified or approved equivalent
Quality Aspects	
Quality of service	Refer to ASTM F3548-21 Section 5.2 for quality of service requirements. Relevant requirements are listed below: *GEN0015 - USSs performing any of the roles identified in this specification shall be implemented and operated under an ISO/IEC 9001-compliant Quality Management System, or equivalent. Service providers may demonstrate equivalent compliance to ISO/IEC 9001 by providing a written attestation that the Service Provider materially complies with ISO/IEC 9001 or has implemented a quality management system that includes the following: • Quality policy

	• Procedures to identify and document key processes and procedures for maintaining, testing, and updating the UTM Services • Process for identifying, investigating, and addressing quality issues and implementing corrective and preventive actions • Criteria for selecting and monitoring suppliers and subcontractors to ensure they meet quality requirements • Plan for fostering continual improvement that includes employee feedback and initiatives to enhance effectiveness and efficiency of processes.
Service verification information	The Network Remote Identification Service including associated DSS interoperability will be verified prior to SW deployment using test baseline identifier TB-ca5d88e
Discovery and Synchronization Service (DSS) pool configuration	DSS configuration process - as defined by InterUSS DSS Pooling instructions on joining an existing pool with new instance. Requirements for confirming DSS pooling integrity Upon joining a pool, a USS hosting a DSS instance shall verify the Dynamic Airspace Representation Identifier matches the published Dynamic Airspace Representation Identifier for the pool. The first instance of the DSS will provide the published Dynamic Airspace Representation Identifier for other pooling entities. USS shall continuously monitor that Dynamic Airspace Representation Identifier for their DSS instances match the DSS pool. Continuously is defined as at least once every 24 hours. For CockroachDB implementation, USS shall notify when pool health metrics are not in bounds within 1 minute of occurrence. In bound health metrics include: • Unavailable Range = 0 Under nominal operating conditions, unavailability is indicative of cluster misconfiguration. • SQL Failure Count = 0 SQL Failure count is a rare but serious indication of a malformed query. Alerting on it in earlier environments helps ensure that an incorrect DSS version update never reaches production. • Active node metric absence for a rolling 5 minutes window. Metric absence indicates that the state of the cluster is unknown and indicates an inability to monitor critical metrics of the system. • Node restarts <= 1 within an hour. Frequent restarts indicate instability.

	Requirements for DSS eviction procedures - Each USS shall periodically cleanup expired Network Remote Identification entities including identification service areas and subscriptions from the DSS pool's shared CockroachDB cluster per the Technical Committee DSS eviction procedures. Requirements for Datastore Connection Pool Sizing - Each USS shall configure the `datastore_max_open_conns` parameter to the value specified by the Technical Committee.														
Safety	Service providers will conduct a hazard analysis to identify and mitigate safety risks in accordance with the process outlined by the GUTMA Safety Task Force Paper " USSP's Safety Support Assessment "														
Performance Requirements															
Nominal service behavior	Refer to the attached Requirements Traceability Matrix, <i>Network Remote Identification Service Compliance Matrix V1.0</i> , for service performance and verification requirements														
Degraded service behavior	Notifications regarding degraded service behavior including DSS degradation will be sent per the terms in the applicable UAS Service Provider Data Sharing and Governance Agreement and Operator-Service Provider agreement. DSS-related service degradation detected per the requirements for confirming pooling integrity and provision of DSS instances are Severity 1 events.														
DSS version and configuration	To participate in the deployment of a DSS pool, a USP must meet the following version and configuration requirements: <table border="1"> <thead> <tr> <th>Setting / Configuration</th><th>Required value</th></tr> </thead> <tbody> <tr> <td>DSS version</td><td>InterUSS DSS v0.23</td></tr> <tr> <td>SCD global lock</td><td>OFF</td></tr> <tr> <td>SCD hash lock</td><td>OFF</td></tr> <tr> <td>Time-based notification index</td><td>ON</td></tr> <tr> <td>CRDB SCD schema version</td><td>3.4.1</td></tr> <tr> <td>CRDB RID schema version</td><td>4.1.0</td></tr> </tbody> </table>	Setting / Configuration	Required value	DSS version	InterUSS DSS v0.23	SCD global lock	OFF	SCD hash lock	OFF	Time-based notification index	ON	CRDB SCD schema version	3.4.1	CRDB RID schema version	4.1.0
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CRDB SCD schema version	3.4.1														
CRDB RID schema version	4.1.0														

DSS infrastructure requirements

To participate in the deployment of a DSS pool, a USP must meet the following minimum infrastructure requirements:

Configuration

- Provide exactly 3 CockroachDB nodes
- Each node must have at least
 - DNS resolution or stable public static IP to enable node discovery
 - Port 26257 availability for inter-node communication
 - CockroachDB version v24.1.3

Provision of DSS instances

These requirements ensure effective minimal provision of DSS instances. It is also recommended that Service Providers adhere to InterUSS recommended sizing and proactively monitor conditions at lower levels to reduce the likelihood of exceeding these requirements.

- USS shall provision DSS instances with the necessary resources to maintain CPU usage of < 90%.
- USS shall notify if CPU usage $\geq 90\%$ for more than 1 consecutive hour.
- USS shall provision DSS instances with the necessary resources to maintain memory usage of < 90%.
- USS shall notify if memory usage $\geq 90\%$ for more than 1 consecutive hour.
- USS shall provision DSS instances with the necessary resources to maintain IOPS in progress < 10.
- USS shall notify if IOPS in progress ≥ 10 for more than 5 consecutive minutes.
- USS shall provision DSS instances with the necessary resources to maintain $\leq 100\text{ms}$ P50 Round Trip Time (RTT) between DSS instances. 100ms is based on the nominal performance for inter cloud RTT times within the continental US as an initial target value.
- USS shall notify if P50 Round Trip Time (RTT) > 100ms more than 5 consecutive minutes.
- For CRDB implementations, the following definitions apply

Requirement	CRDB metric per CRDB node
CPU Usage	sys.cpu.combined.percent-normalized
Memory	sys.rss
IOPS in progress	sys.host.disk.iopsinprogress
Storage	capacity.used

	P50 Round Trip Time (RTT)	round-trip-latency (50 Percentile)
	As of the reference date of this SDD, the following configuration generally meets the above Provision of DSS instances requirements ○ CPU: At least 4 cores; 8+ cores recommended, with support for virtualization and containerization. ○ RAM: At least 8GB of memory; 16GB+ recommended. ○ Storage: SSD-based storage with at least 100GB available per node (expandable based on dataset size and replication needs). ■ Minimum IO/s 3000 R/W ■ Minimum throughput 125 MB/s R/W ■ Maximum average latency 5ms R/W ■ Maximum P99 R/W latency 10ms ○ Network: High-speed (10+Gbps) low-latency network with a stable public IP address ○ Accurate system time using NTP synchronization from Google's Public NTP source	